

Its Not That Deep – Or Is It? Determining Material Density Using Cosmic Ray Muons

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Abstract

Muon tomography, a newly developing imaging technique that uses muon particles from cosmic rays to 'x-ray' objects, avoids the considerable costs and safety risks of traditional imaging methods.

However, existing research has relied on expensive, large-scale apparatuses, limiting accessibility and widespread adoption.

This study investigates whether inexpensive, handheld muon detectors can accurately determine object density, an essential task in imaging.

We recorded the number of muons detected over time while objects of equal volume but varying densities were placed atop the detectors.

Higher-density objects blocked more cosmic ray muons from reaching the detectors. By fitting a model to the data, we established a relationship between muon attenuation and object density, demonstrating the feasibility of using inexpensive detectors for density estimation.

This study implements the fundamental process underlying muography, increasing the accessibility of this promising technique.

Methods: Theoretical Basis

Muon particles are created by cosmic rays in the atmosphere that collide with atomic nuclei. These collisions produce particles with very short life spans, known as pions and kaons, which ultimately decay into muons. Although muons have a relatively short proper lifespan, 2.2 microseconds, they move at relativistic speeds. Time dilation, an effect of special relativity permits muons to live long enough in our standing reference frame to reach the ground level, and ultimately allow muons to be detected by humans.

The energy loss rate, also referred to as dE/dx or stopping power in other academic papers relating to muons is a figure representing approximately how much energy a particle loses as it passes through material, measured in units of $\text{MeV cm}^2/\text{g}$. For a typical material with a density of 1.0 g/cm^3 , a minimum ionizing muon loses approximately 2.2 MeV per g/cm^2 traversed (Axani et al.).

This known trend in energy loss was used to our advantage in defining a concrete method to detect muons. If a muon penetrates a material that's thick enough (path length) or dense enough (density), it will lose energy along its path. The tendency of muons to deposit energy as it travels through material is integral to the functionality of a muon detector based in a scintillator and photomultiplier design, which is the type of detector we built.

Methods: Designing the Muon Detector

The detector we built was modeled after the MIT CosmicWatch project's design. Its detection is based on plastic scintillation with a SiPM for photodetection and signal processing through an Arduino Nano.

The plastic scintillator is the core of the detector. When a high energy muon traverses the scintillator, it excites the molecules in the plastic, emitting light when it strikes. The light that is emitted is near-visible—a process known as scintillation. However, the photon emitted is typically weak, so maximizing the photon collection is imperative for reliable detection.

First, the photons are read by a silicon photomultiplier (SiPM), which is placed right against one of the scintillator faces. The SiPM is a highly sensitive amplifying photodetector composed of an avalanche of photodiodes operating in what's known as Geiger mode (Axani et al.). This just means that the diode is configured so that a single electron-hole pair, which would be created by the absorption of a photon, would trigger a massive avalanche effect that would be much greater than the original photon produced.

Each photon that strikes the SiPM will create a cascade effect of elections that produces a voltage, which is then read by the Arduino and processed by us. The Arduino provides us a table of data including the timestamp of the event, event count, and voltages detected (Figure 1). An OLED screen was used to display the time elapsed, the raw count, and the rate of detection (Figure 2A).

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### CosmicWatch: The Desktop Muon Detector
### Questions? saxani@mit.edu
### Comp_date Comp_time Event Ardn_time[ms] ADC[0-1023] SiPM[mV] Deadtime[ms] Temp[C] Name
#####
Device ID(s): Sanjai
2025-04-21 13:11:15.665576 1 3468 88 20.96 0 22.83 Sanjai
2025-04-21 13:11:19.118100 2 6909 57 16.93 5 22.83 Sanjai
2025-04-21 13:11:23.178869 3 10952 449 112.05 10 22.83 Sanjai
2025-04-21 13:11:24.759419 4 12529 71 19.06 15 22.83 Sanjai
2025-04-21 13:11:25.398715 5 13166 89 21.05 20 23.05 Sanjai
2025-04-21 13:11:26.267834 6 14031 103 22.23 25 23.05 Sanjai
2025-04-21 13:11:26.873473 7 14636 345 64.98 30 23.05 Sanjai
2025-04-21 13:11:34.402420 8 22139 207 35.26 35 22.62 Sanjai
2025-04-21 13:11:40.186956 9 27905 51 15.86 40 22.94 Sanjai
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Figure 1: Arduino collected data detailing event date, time, event count, ADC read, SiPM voltage, length of event, temperature read, and the name of our device.

In order to further maximize photon collection, we wrap the scintillator in aluminum foil. The foil acts as a reflective surface so that photons will be reflected internally, toward the SiPM in order to further amplify any signal that may be produced. We also encase the entire apparatus in electrical tape because we don't want the outside light to interfere or create false signals. All pieces were hand soldered (Figure 2B).

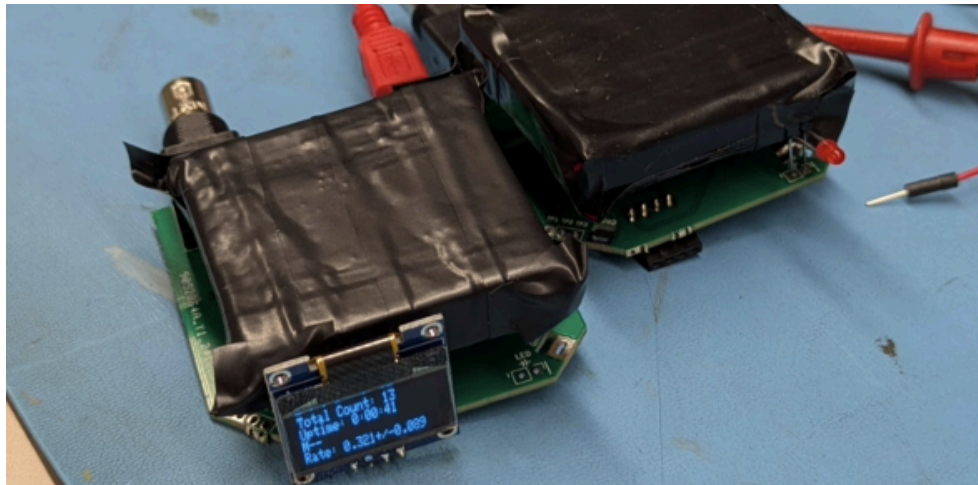
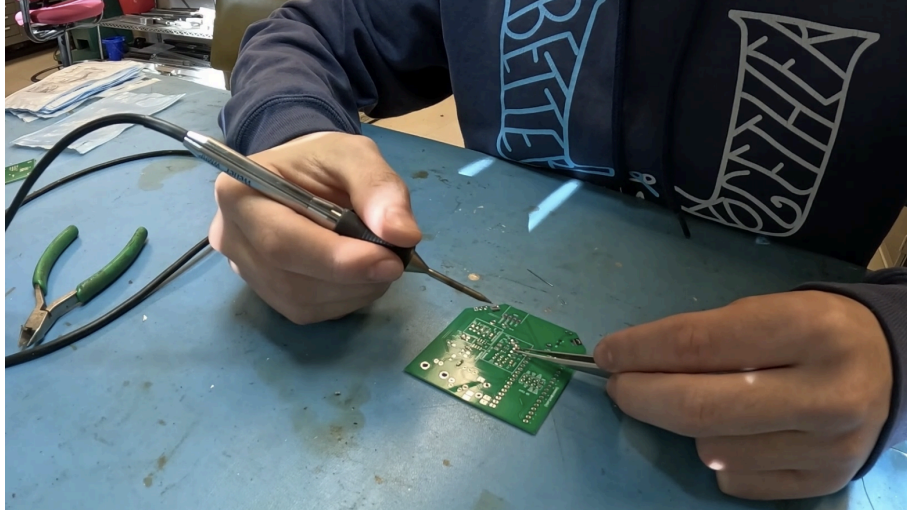


Figure 2A: OLED screen displaying muon count, elapsed time, and detection rate during live operation.



Figures 2B: Soldering a Resistor onto the main CPB of the muon detector

Testing Process and Procedures of Experimentation

Density Experiment: Procedures

After constructing a muon detector, we based our validation of the detector on the principle that muons lose a substantial amount of their energy as they pass through materials. Our goal was to test whether the detector was truly detecting muons. The attenuation of a muon particle's energy is determined by the stopping power, so as the amount of material a muon has to pass through before reaching the detector increases, the number of muons that actually reach the detector should decrease. Only muons with sufficient energy to penetrate the added material and still trigger the scintillation process should be detected. This principle was the basis of our method of testing our detector. We were aiming to discover a predictable pattern in attenuation rate versus the thickness of a barrier the muons have to pass through.

We conducted two separate experiments in order to test whether the behavior of our detector was consistent with prior research. In the first experiment, we stacked plates of Steel (7.9 g/cm^3 density) and Aluminum (2.7 g/cm^3), creating a 11.43 cm barrier between the detector and the atmosphere. Additionally, we ran a control with no barrier. By varying density, we varied the stopping power of the material on top of the detector without varying how much material a particle would have to pass through (Niederleithinger et al.). Our hypothesis for this density experiment was that an increase in density creates an increase in attenuation and therefore a decrease in muon count.

For both experiments, data were collected by placing the detector on a table in the second floor of the Irvine Valley College B300 building. We also used a steel tube to encase the detector to minimize edge effects caused by muons striking the scintillator block from the sides (Figure 3). The device was powered on and allowed to run continuously, with the live detection events recorded via the Arduino data feed. Finally, we took the raw counts and divided them by the elapsed runtime to calculate the average count rate. Additionally, we normalized the rate by the surface area of the detector, which is 25 cm^2 , to get the rate per area of the detector.

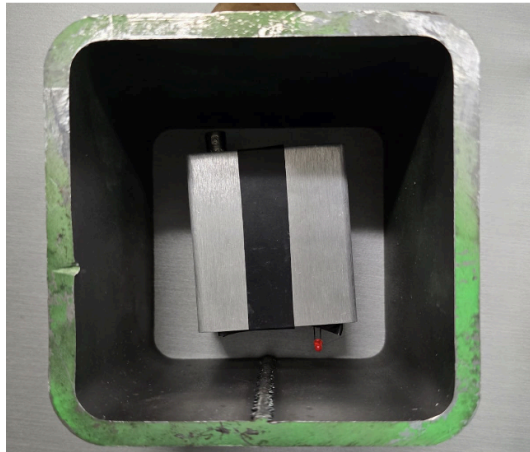


Figure 3: Muon detector encased in a steel tube to prevent edge effects.

Density Experiment: Results

This experiment aimed to test whether increasing the density of the material above our CosmicWatch designed detector would reduce muon flux and remain consistent with the expected attenuation behavior associated with professional muon detectors. Our results show that there was an observable decrease in the rate of detection as we increased the density of matter above, suggesting a consistent attenuation effect (Figure 4).

Steel vs Aluminum Thickness: 11.43 cm Scintillator: 25 cm ²				
	Time elapsed (min)	Count	Rate (n/min)	Rate per area (n/min/cm ²)
air	24	420	17.5	0.7
Aluminum 2.7g/cm ³	91	1431	15.73	0.629
Steel 7.9g/cm ³	67	986	14.73	0.5894

Figure 4: Muon detection rates (with normalization per cm^2 of the detector surface area) for aluminum, steel, and air at constant thickness, 11.43 cm. Higher density correlates with lowered muon flux.

Our hypothesis, that increasing the density of the matter will decrease muon flux was overall supported by the results of the experiment. The muon flux without a barrier is also somewhat

close to the literature value, $1 \text{ muon/cm}^2/\text{min}$ (Gadey et al.). Slight variations in count rates may be due to environmental fluctuations, however the overall trend remains clear. Our main goal going forward would be to run more trials to ensure this trend is consistent.

These results ultimately demonstrate that it's possible to create inexpensive and small-scale detectors that are still reliable in detecting muon attenuation patterns, creating a more feasible option for undergrad research in muography and muon tomography.

Thickness of Barrier Experiment: Procedures

The second experiment approached the attenuation patterns from a different perspective. Instead of varying density, we tested the same material but barrier thickness was incrementally increased. We used Steel, with a density of 7.9 g/cm^3 as our test material and started from 0 cm, increasing to 11.43 cm with increments of 3.81 cm (1.5 inches). By increasing the amount of material sitting on top of the detector, we expected a decrease in muon flux because while the stopping power (dE/dx) is constant, total energy loss increases, and more opportunity to stop a muon.

Our hypothesis for this thickness of barrier experiment was that an increase in material creates a quantifiable and significant decrease in muon flux. This experiment complements the density experiment because there are really two factors to the full stopping of a muon particle: path length and stopping power. Muon flux should be sensitive to both in order for us to validate the detectors performance.

Thickness of Barrier Experiment: Results

The goal of this experiment was to challenge whether or not our detector would behave consistently with the expected attenuation trends associated with an increased path length from the muon to the detector. Our results yield an quantifiable trend detailing the decrease in muon flux as we increase the amount of steel that a muon must pass through in order to reach the scintillating block (Figure 5A).

Varying amounts of Steel

Density: 7.9 g/cm^3

Scintillator: 25 cm^2

Thickness of Steel (cm)	Detections/min	Rate/Area of Scintillator
0	17.5	0.7
3.81	16.96	0.6784
7.62	15.73	0.6292
11.43	14.73	0.5894

Figure 5A: Muon detection rates for steel (7.9 g/cm^3) with varying thickness. Higher material thickness corresponds to lower muon flux.

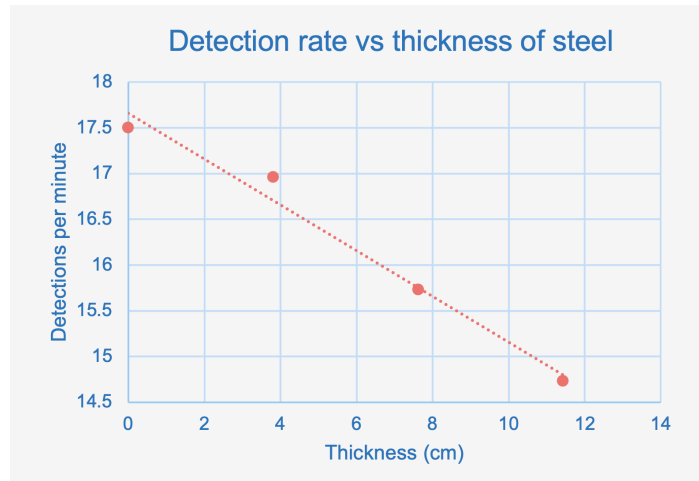


Figure 5B: Graph of muon detection rate (N/min) versus steel barrier thickness. Linear decline trendline suggests attenuation and thickness have a proportional negative relationship, as hypothesized.

These results support our hypothesis that increasing the thickness of a barrier on top of the detector would decrease the amount of muons detected over time. It produced a linear trend with a slope of -0.25, which represents the change detection rate as a function of thickness stacked. The negative trend is what we expect from past studies. We also plan to run this experiment several more times and with new detectors that are under development. However at this early stage in the project, the predictable trend and consistency of this detector again support the feasibility of employing inexpensive equipment in muon research going forward.

Conclusion

Our muon detector, modeled after MIT's CosmicWatch Project produced an observable and reproducible trend that was consistent with the expected attenuation behavior discovered in past muography studies. Our findings not only demonstrate the validity of our efforts to create a working muon detector, but also demonstrate that particle physics research can be made accessible to undergrad physicists and engineers.

Currently, larger scale research is being done on the development of muon imaging techniques, or the field known as muon tomography, and these developments have wide applications in archaeology, materials science, and civil engineering (Bonechi et al., Zenoni et al.). However, these studies often rely on large-scale, high-cost muon detection apparatuses and it may be worth exploring just how large the scope of these small, handheld muon detectors' capabilities expand.

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